Cybernetics, Automata Studies, and the Dartmouth Conference on Artificial Intelligence

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The Dartmouth Summer Research Project on Artificial Intelligence, held at Datrmouth College in 1956, is regarded as the official ''birthplace'' of AI. This article draws on unpublished archives to shed new light on the origins of the conference and the complex relationships between cybernetics, automata studies, and AI in the 1950s.

The Dartmouth Summer Research Project on Artificial Intelligence, held at Dartmouth College in Hanover, New Hampshire, in the summer of 1956, is generally regarded by historians and computer scientists as the ''birthplace'' of AI.[1] Pamela McCorduck argued that two events mark its significance: the debut of the ''new paradigm'' of symbolic information processing, represented by Allen Newell and Herbert Simon's Logic Theorist computer program, and the ''social patterns'' set by participants who founded the early AI programs at the Massachusetts Institute of Technology (MIT), Stanford University, Carnegie Mellon University, and the Stanford Research Institute.[2] These institutions promoted the approach that came to be called *symbolic AI*, rather than the earlier method of brain modeling, a neurophysiological approach associated with cybernetics and neural nets. Participants at Dartmouth have disagreed about how to characterize the conference's significance. Their interpretations range from viewing it as a milestone in the field of machine intelligence (Oliver Selfridge), to being deeply disappointed that it was not a collective research enterprise (John McCarthy), to seeing Dartmouth as a ''piece of AI mythology'' (Newell).[3]

My aim here is not to adjudicate these varying interpretations, but to draw on unpublished archives to shed new light on the conference as well as the historical relationships among cybernetics, automata studies, and AI. I explore the disciplinary boundary work done by researchers and the Rockefeller foundation, which funded the conference, to separate brain modeling from symbolic processing during the editing of the volume *Automata Studies* (1956) by McCarthy and Claude Shannon and the planning, funding, and conduct of the Dartmouth conference. I argue that we gain a better understanding of the early history of AI, and of cybernetics, by placing *Automata Studies* and the Dartmouth conference in the context of the variety of research done under the rubric of cybernetics at the time.

Cybernetics

One of the chief meanings of cybernetics in the 1950s, which MIT mathematician Norbert Wiener defined in *Cybernetics* (1948) as the science of control and communication in the animal and machine, centered on machine intelligence.[4] In the book, Wiener dealt with the analogy between the new digital electronic computers and the supposed on/off characteristic of neurons in the human brain, which neurophysiologist Warren McCulloch and mathematician Walter Pitts had formally established in a 1943 paper on the computability of neural nets. The analogy was explored extensively at the Macy Foundation conferences on cybernetics chaired by McCulloch and held from 1946 to 1953.[5] Wiener discussed the possibility of a chess-playing computer program in *Cybernetics*;[6] and several cybernetics robots, such as Wiener's moth/bedbug, W. Ross Ashby's homeostat, and W. Grey Walter's tortoises, were built in the early 1950s to simulate psychological and neurological functions. (Ashby was a British psychiatrist and pathologist, and Grey Walter a British neurophysiologist.[7])

Several participants at the Dartmouth conference had been influenced by cybernetics. Selfridge had proofread Wiener's *Cybernetics* as a graduate student at MIT, Marvin Minsky had been ''enamored'' with McCulloch's work on neural nets, and Simon had relied upon Ashby's research, instead of Wiener's, to develop the idea that humans and computers were adaptive information processors.[8] Later recollections also note the importance of cybernetics in the early history of AI. Minsky claimed that the advent of the digital computer in the 1950s ''made it practical to be much more ambitious'' in designing intelligent machines. As a result, cybernetics divided, in my view, into three chief avenues:'' minimal self-organizing systems (represented by Ashby), simulating human thought (represented by Newell and Simon), and AI (represented by Minsky and McCarthy).[9] Newell recalled that after the mid-1950s, cybernetics split with AI over the issues of ''symbolic versus continuous systems'' and ''psychology versus neurophysiology.'' The break was not abrupt:

> Through the early 1960s, all the researchers concerned with mechanistic approaches to mental functions knew about each other's work and attended the same conferences. It was one big, somewhat chaotic, scientific happening.[10]

Many historians have also seen cybernetics as an origin point for early AI.[11]

Shannon, a mathematician at Bell Telephone laboratories who founded the American school of information theory,[12] also had a strong interest in cybernetics and automata. He wrote early articles on programming computers to play chess, corresponded on the subject with Wiener (his former professor at MIT), described his work in information theory to McCulloch in 1949 under the rubric of '''Cybernetic' questions,'' belonged to a cybernetics study group in New York City in 1950, and built a relay-logic, maze-running ''mouse.''[13] While attending three Macy conferences on cybernetics in the early 1950s, where he presented research on both information theory and automata, he got to know McCulloch; Pitts; Grey Walter; engineer Julian Bigelow, who coauthored a founding paper on cybernetics with Wiener and was John von Neumann's chief engineer on the Institute for Advanced Study (IAS) computer project at Princeton; and Donald MacKay, a founder of the British school of information theory who did research on automata.[14]

Although Shannon concentrated on information theory, he maintained his interest in cybernetics to the extent of writing the original article on the subject for the *Encyclopedia Britannica*. In that article, published in 1957, he said that cybernetics ''overlaps'' several fields, including information theory. Wiener had referred the Britannica editor to Shannon to write the entry in his place.[15]

Automata Studies

Shannon's interest in cybernetics helps us understand McCarthy's experience coediting *Automata Studies* with Shannon in the early 1950s, which led directly to McCarthy organizing the 1956 Dartmouth conference. McCarthy, who had worked under Shannon at Bell Labs in the summer of 1952 after finishing his PhD in mathematics at Princeton, recalled that he considered Shannon's title for the proposed book to be too conservative and thus used the new term ''artificial intelligence'' when he proposed a summer-long conference devoted to the subject. Correspondence related to the automata studies project gives us a glimpse of these tensions, but it also indicates that Shannon held a broader view of automata studies than did McCarthy.[16]

In line with Shannon's interest in cybernetics, in 1953 he and McCarthy invited several researchers who had attended the Macy conferences on cybernetics to contribute to the proposed volume, including Wiener, MacKay, Ashby, Grey Walter, and von Neumann. Familiar with Shannon's theory of information, von Neumann had begun a series of research on automata theory based on the McCulloch-Pitts logical model of neural nets and Turing machine theory.[17] Shannon and McCarthy invited other prominent researchers in the theory of automata and neural nets, including Alan Turing himself, whom Shannon had met during World War II and had visited in England;[18] S.C. Kleene at the University of Wisconsin–Madison, who had written a paper on neural nets while at RAND in 1951; and Albert Uttley, who was designing conditional-probability neural nets for pattern recognition for the British military.[19] Minsky, who had also been hired to work with Shannon at Bell Labs in the summer of 1952,[20] agreed to submit something from his doctoral dissertation on neural nets and brain modeling that he was working on at Princeton. Wiener thought

highly of Minsky's research, saying in a 1954 recommendation letter that it went "considerably further than the best work to date of such men as Ashby and Grey Walter."[21]

The British researchers Turing, MacKay, Ashby, Grey Walter, and Uttley were regular members of the Ratio Club, a group of neurophysiologists and mathematicians who had worked on cybernetics before Wiener published his book of that title. They held their first meeting on the occasion of McCulloch's visit to England in 1949.[22] In 1953, Turing referred directly to the cybernetic aspect of the automata-studies enterprise in his response to the invitation from Shannon and McCarthy to contribute a paper to the volume. Turing had to beg off because he had "been working on something quite different for the last two years; viz. the mathematics of morphogenesis, though I expect to get back to cybernetics very shortly."[23]

McCarthy played a substantial role at the beginning of the project. Prospective authors corresponded with him in care of "Automata Studies, Fine Hall" at Princeton. The young McCarthy was not reticent in stating his views on how the papers fit into his vision of the field. MacKay's paper discussed how automata could symbolically represent the external environment in a self-organizing, adaptive manner and proposed an analog electronic circuit, operating electromagnetic gates, to illustrate the principle of statistical negative feedback involved in his cybernetic scheme. McCarthy told Shannon,

> I think we should include it [the paper] with whatever changes he wishes to make. I don't think that the eventual solution to this problem will be an extension of his ideas, but I have no very precise grounds for this opinion.[24]

After criticizing MacKay for making questionable assumptions about sensory transformations based on the McCulloch-Pitts neural-net model, McCarthy concluded, "Nevertheless, I think the paper has a number of good ideas and also represents a point of view which cannot be dismissed as unfruitful this early in the game."[24]

McCarthy had more trouble with MacKay's colleague Ross Ashby. Having gained some prominence with the publication of *Design for a Brain* (1952), which presented his homeostat and a theory of adaptive behavior, Ashby interpreted the invitation from Shannon and McCarthy as being an automatic acceptance for his paper, "Design of an Intelligence Amplifier." Ashby equated "intelligence" with the capacity to make appropriate selections and extended his work on the homeostat, an electromechanical device that adapted to changes in its "environment" through a process of self-organization, by randomly selecting states that would eventually lead to stability (homeostasis). Now, he presented the general principles for designing a "selection amplifier," which would speed up the process and reach solutions not contemplated by the designer, resulting in an "amplification" of intelligence.[25] Ashby told McCarthy,

John McCarthy played a substantial role at the beginning of the *Automata Studies* project.

> I am afraid I cannot consent to having it [his paper] "refereed," as it is on a subject to which I have devoted a great deal of work, and on which I can now speak with some authority. Frankly, I know of no referee to whose opinion I would defer (though I would, of course, be prepared to answer specific objections).[26]

In contrast, MacKay welcomed referee reports on his paper. McCarthy and Shannon ended up accepting the Ashby and MacKay papers without refereeing them.[27] The experience probably did nothing to improve McCarthy's opinion of Ashby, the dean of the English cyberneticians.

Shannon and McCarthy divided the articles into three categories: finite automata, Turing machines, and synthesis of automata. The categories were a subset of the subject headings Shannon had used in a comprehensive review of computers and automata research published in 1953. They were also part of his broad view, expressed in 1958, of automata theory as a recent, ill-defined "interdisciplinary science," which spanned symbolic logic and Turing machine theory, nonnumerical (symbolic) programming of computers, and neural nets.[28] The only category missing from the automata studies

volume was that of symbol manipulation by computers. Cybernetics was well represented; almost half the papers (six of 13) were on the theory and practice of neural nets. The British cyberneticians made up the entire section on the synthesis of automata.

All in all, McCarthy was disappointed. He wrote Shannon in mid-1954, ''The collection as a whole does not represent great progress but is certainly representative of current thought.'' McCarthy's expectations were high. He thought the papers by the Ratio Club group (Ashby, MacKay, and Uttley) ''contribute heuristic ideas of some value but none lead directly to a solution of the problem of thinking automata.'' The papers by Kleene and James Culbertson on neural networks ''solve problems of mainly formal interest.'' McCarthy thought his own paper on Turing machines introduced ''concepts which may prove useful but does not get far off the ground.'' Shannon's paper was a ''short step in the direction of the theory of computing rate [of Turing machines].'' Von Neumann's paper on probabilistic, multiplexed, neural-net automata made from unreliable components got the best mark: ''An important result, perhaps slightly off the main track.''[29] Minsky's paper had not yet arrived for comment.[30] Not surprisingly, Shannon, who had praised Ashby's homeostat in print ''as a basis for learning machines and brain models,'' had a better opinion of the collected papers.[31]

The tensions over editing *Automata Studies* recalled by McCarthy surface in a letter he wrote Shannon in late 1954. McCarthy had written a draft of a preface to the volume. ''It consists mainly of a point of view on how the various lines of investigation represented by the included papers may contribute to the eventual design of intelligent machines.'' The sentence is crossed out by a wavy line, perhaps by Shannon, in the original letter deposited at the Shannon Papers at the Library of Congress. McCarthy also asked some pointed questions of Shannon:

1. Do you agree that the book should be regarded as a step in the direction of the design of intelligent automata? Or is this too presumptuous? Might it surprise the contributors?
2. What about the title ''Towards Intelligent Automata''? Is it too bombastic? Others?[32]

Apparently Shannon did think McCarthy's interpretation of the volume was ''too presumptuous'' and the proposed title ''too bombastic.'' The preface to the published volume de-emphasized the goal of designing ''intelligent automata'' and spoke, instead, in terms of the ''analytic and synthetic problems'' of automata.[33] The title remained the staid *Automata Studies*. Yet Shannon sympathized with McCarthy's goal. He wrote his former high school science teacher in 1952, ''My fondest dream is to someday build a machine that really thinks, learns, communicates with humans and manipulates its environment in a fairly sophisticated way.''[34] Shannon also approved the publication of papers by the Ratio Club group, representing the British cybernetics movement, on design principles of intelligent automata.

McCarthy was not adverse to mentioning cybernetics in the book's preface. In the letter to Shannon, he listed cybernetics as one of six areas toward which the preface could, perhaps, make ''brief bows'' (the others were mathematical logic, statistics, communication theory, neurophysiology, and telephone engineering).[35] The published preface, signed by McCarthy and Shannon, bowed toward all these areas or to their practitioners by name—except for cybernetics. The archives do not indicate who deleted the reference to cybernetics.

Rockefeller Funding of the Dartmouth Seminar

The period from completing *Automata Studies* to approaching the Rockefeller Foundation to fund a summer seminar on AI was short. Upon moving to Dartmouth College in February 1955 to take up an assistant professorship in mathematics, McCarthy sent Shannon the final version of his paper for the volume and said he was now free to do additional editing on it and to visit Shannon at Bell Labs.[36] In early April, McCarthy had an interview with Warren Weaver, director of the Natural Sciences division of the Rockefeller Foundation, about the possibility of funding a six-week summer seminar at Dartmouth for about 10 participants to work together on brain theory and automata. Weaver recorded in his program officer's diary, which was circulated within the foundation, that McCarthy was confident that Shannon and Nathaniel Rochester at IBM, an architect of the IBM 701, would attend and that their employers would pay their salaries while they were at Dartmouth. Weaver said he would think it over.[37]

Weaver came to have mixed feelings about the project and told McCarthy that Robert Morison, director of the Biological and Medical Research division of the foundation, should handle the proposal,[38] probably because it dealt with brain models. Unable to attend the scheduled lunch in mid-June between Morison, McCarthy, and Shannon, whom McCarthy had enlisted to promote the project, Weaver expressed his opinion on the matter to Morison. An applied mathematician who had directed the foundation's funding of early computer projects such as the Rockefeller Differential Analyzer,[39] Weaver felt it was "rather a 'personal' project, in the sense that McC. [McCarthy] and two or three other people would enjoy spending some time together, talking about various aspects of information theory" at someone else's expense. Weaver did not think the foundation should support it on that basis. On the other hand, Weaver had the highest regard for Shannon, whose theory of information he had popularized in the jointly authored *Mathematical Theory of Communication* (1949) and thus worried about a conflict of interest. "I think it is on the whole rather better that I am not going to be at the luncheon. My personal connection with the subject, especially since Claude Shannon is now going to be present, tends to remove the discussion from the objective atmosphere in which we ought to approach it."[40]

Shannon's presence seems to have inclined Morison toward the proposal. Morison recorded in his diary that McCarthy did almost all the talking at the meeting, with Shannon only answering questions put to him. "McCarthy strikes one as enthusiastic and probably quite able in mathematics," Morison wrote, "but young and a bit naive." Participants suggested for the seminar were McCarthy, Shannon, Rochester, Minsky, von Neumann, Selfridge, and psychologist Donald Hebb, who was well known for his neural net theory of learning. Having funded, in conjunction with Weaver's division the mathematical and neurophysiological research of Wiener and Arturo Rosenblueth in the mid-1940s—which led Wiener to write *Cybernetics*—Morison apparently wanted to include someone from Wiener's group at MIT in the Dartmouth conference. Morison said that Selfridge, who was now a group leader at MIT's Lincoln Laboratory, "apparently strikes M. [McCarthy] and S. [Shannon] as the soundest and most amenable to group activity of the various younger men who have been associated with Norbert Wiener." Morison was alluding to the difficulties Wiener and Rosenblueth had in obtaining consistent work from the erratic, but brilliant Walter Pitts during the Rockefeller-funded research.[41] Minsky might also have recommended Selfridge, with whom he had organized a session on machine intelligence at the ACM's Western Joint Computer Conference in 1955.[42]

Two other prominent cybernetics researchers were considered at the luncheon. Despite their difficulties working with Ross Ashby during the *Automata Studies* venture, McCarthy and Shannon put forward Ashby as "the most attractive foreign name." MacKay, whose research on information theory and analog automata had been funded by the Rockefeller Foundation since 1951 and whose paper was accepted for *Automata Studies*,[43] did not fare as well. Morison recorded that "S. [Shannon] worries a good deal about MacKay's interest in theological metaphysics." Although MacKay, who also wrote on the philosophy of science, maintained that his research in neuroscience and his Christian faith were complementary, not antagonistic, Shannon was apparently concerned that MacKay's religious views would create frictions in an informal six-week seminar. Morison suggested including two psychologists, one of whom was Hans Lukas Teuber at New York University, who had coedited the proceedings of the Macy conferences on cybernetics. Morison thought both men had "enough breadth of education and curiosity to have informed themselves, at least fairly well, about cybernetics and mathematical speculations"[44]—one of the few direct references to cybernetics in regard to the Dartmouth conference.

Encouraged by the luncheon, McCarthy organized a committee to prepare a proposal to fund the conference. McCarthy, Shannon, Minsky, and Rochester, with whom McCarthy had worked at IBM in the summer of 1955,[45] finished their research sections of the proposal by late August, and McCarthy submitted it in early September 1955.[46] As Paul Edwards argued, the "still-nascent split between the computer-brain and computer-mind metaphors already appears clearly" in the proposal.[47] Yet the proposal includes both approaches as different aspects of a common enterprise utilizing the computer as a research tool. The introduction, probably written by McCarthy, describes the "artificial intelligence

problem'' under seven headings: Automatic Computers, How Can a Computer Be Programmed to Use a Language, Neuron Nets, Theory of the Size of a Calculation, Self-Improvement, Abstractions, and Randomness and Creativity. The section on computer languages, McCarthy's emergent research area, was clearly in the camp of what came to be known as symbolic AI. The section on neural nets mentioned the research of McCulloch and Pitts; Uttley, who had contributed two articles to *Automata Studies*; and two of the organizers, Minsky and Rochester.[48]

The research statements of the organizers reveal the ''nascent split'' in more detail. Three of the four organizers were on the brain-modeling side. Shannon proposed to work on two topics: the ''Application of information theory concepts to computing machines and brain models'' and the ''matched environment—brain model approach to automata.'' The first project extended his previous work on computing with unreliable components, which could apply to computers and human brains. The second project would focus on mathematizing the model of the brain's ''environment.'' Shannon concluded by saying he proposed to ''work up through a series of stages in the direction of these advanced activities [of human thought].'' Minsky hoped to extend his dissertation research and complete a model of a trial-and-error learning neural network that could be programmed on a computer. Rochester wanted to continue the research of his group at IBM on programming neural nets on a computer to exhibit originality when faced with a new problem by inserting randomness into this type of brain modeling. Only McCarthy leaned toward symbolic processing by proposing to ''study the relation of language to intelligence.'' He criticized the brain-model approach: ''It seems clear that the direct application of trial-and-error methods to the relation between sensory data and motor activity will not lead to any very complicated behavior.'' He thought trial-and-error methods should be applied at a ''higher level of abstraction,'' for example, with an English-like ''artificial language which a computer can be programmed to use on problems requiring conjecture and self-reference.''[49]

The proposal did not sit well with Morison at the Rockefeller Foundation. He told McCarthy in late September that the ''proposal is an unusual one and does not fall easily into our program [in biological and medical research] so I am afraid it will take us a little time before coming to a decision.''[50] Not having heard from Morison by early November, McCarthy asked Shannon to approach Weaver and determine its status.[51] In late November, Morison finally told McCarthy that the foundation could not grant the requested funding of $13,500 for the seminar, but could support a smaller effort at $7,500. Morison wrote,

> I hope you won't feel that we are being overcautious, but the general feeling here is that this new field of mathematical models for thought, though very challenging for the long run, is still difficult to grasp very clearly. This suggests a modest gamble for exploring a new approach, but there is a great deal of hesitancy about risking any very substantial amount at this stage.[52]

McCarthy and the other organizers acquiesced and said they could get by with $7,500.[53] In deciding whom to invite, McCarthy made what turned out to be a fateful trip to visit Newell and Simon at the Carnegie Institute of Technology (CIT, now Carnegie Mellon University) in February 1956. By that date Newell and Simon, working at RAND's Systems Research Laboratory, had essentially completed the Logic Theorist, one of the earliest software programs in symbolic AI. Newell recalls that he was inspired to work in this direction after hearing Selfridge give a talk on a pattern-recognition program at RAND in 1954.[54]

During his visit to CIT, McCarthy showed Newell and Simon the Rockefeller proposal for the Dartmouth seminar. Simon wrote Shannon after the visit, reintroducing himself by reminding Shannon of a RAND report Simon had sent him on a chess-playing program and enclosing a copy of an article of his that bore on the research Shannon proposed to do at Dartmouth. Then Simon took the liberty of critiquing Shannon's proposed research. In light of the work he had been doing with Newell, Simon said Shannon's two separate projects—on using information theory to analyze computing with unreliable components, and modeling the brain's simplest environments—were actually one project. ''Because of the rapid progress we have made in the last two months,'' Simon concluded, ''we are inclined to reverse the strategy expressed in the last sentence of your proposal—we believe that we can start with some of the most advanced human activities—i.e., proving theorems—and work

back to the 'simplest'—e.g., learning nonsense syllables.'' Newell planned to ''spend part of the summer at Dartmouth, and I shall plan at least a week there at the beginning, if this is agreeable to the 'proposers.'''[55] McCarthy recommended Newell and Simon to the other proposers, and Newell and Simon wrote up their plans for doing research at Dartmouth.[56]

Shannon was slow to respond to Simon's rather challenging letter. Two months later, in late April, Shannon wrote to Simon, apologized for the delay, and admitted he had not done much research on learning machines in the past two years. Shannon was on the committee of a master's student, Trenchard More, at MIT, where Shannon had recently been appointed a visiting professor. Shannon said that More ''has been developing a program quite similar to that you speak of for proving theorems in the propositional calculus.'' Like the program developed by Newell and Simon, More's could be simulated by hand. Perhaps to mollify Newell and Simon, Shannon included an announcement of the call for papers for the Symposium on Information Theory, sponsored by the Institute of Radio Engineers' (IRE) Professional Group on Information Theory, to be held at MIT in September 1956 and suggested ''that perhaps some of the work you or Newell have been doing would be suitable for presentation'' in a special session on learning machines.[57]

The invitation pleased Newell and Simon, but the implication that a master's student of Shannon's was doing work comparable to theirs irked them enough for Newell to bring up the letter with some passion in an interview conducted 35 years later:

> So we have a letter [from Shannon]. And in this letter there is some stuff about a guy by the name of Trench Moore [sic], Trench was a student of Shannon's, and in fact Shannon thought that what Trench was doing was infinitely further ahead than what we were doing. So there's this funny little exchange and counter exchange and so forth that relates to what Trench is doing and so forth.[58]

By May 1956, McCarthy had formalized the arrangements for the seminar, which was to be held from mid-June to mid-August at Dartmouth College. McCarthy told the Rockefeller Foundation that the participants for the full period would be himself, Minsky, MacKay, Ray Solomonoff, a colleague of Minsky's at MIT, John Holland from Rochester's group at IBM, and Jacob Bigelow. Shannon, Rochester, and Selfridge planned to attend two weeks at the beginning of the summer and two weeks at the end. Newell and Simon would attend for two weeks at the beginning. To help make up the Rockefeller Foundation's budget cut, IBM and Lincoln Labs would pay the salaries of their employees while they attended the seminar. McCarthy hoped the conference would focus on his area of research, telling Morison, ''At present it looks as though we shall concentrate on a problem of devising a way of programming a calculator to form concepts and to form generalizations.''[59]

The conference did not work out as planned. Several of the participants listed by McCarthy did not attend. MacKay, who was probably added at the request of the Rockefeller Foundation, and whose travel from Britain the foundation was funding from another account, cancelled at the last moment. He decided to stay with his wife who was ordered by her doctor to remain in bed following a late delivery of their first child.[60] That removed an outspoken advocate of the cybernetic analog approach to brain modeling from the seminar. Holland at IBM decided not to attend because of previous commitments, a decision he later regretted.[61] Shannon did not make it either and probably sent his student Trench More in his place, most likely to continue the conversation with Newell and Simon. Bigelow does not appear in accounts of the Dartmouth seminar. New attendees included Herb Gerlertner and Arthur Samuel, who wrote an early checkers program.

The 10 official participants were thus Gerlertner, McCarthy, Minsky, More, Newell, Rochester, Samuel, Selfridge, Simon, and Solomonoff. Alex Bernstein from IBM was a visitor.[62] Bernard Widrow (who received his PhD from MIT that summer) and a colleague heard about the conference, drove from Cambridge to Dartmouth on their own initiative, and were welcomed by the group. Widrow recalls that the group also included W.A. Clark and B.G. Farley from Lincoln Labs, who were probably visitors. Widrow was ''fascinated by the things I heard [at Dartmouth] about artificial intelligence,'' which shaped his later research on adaptive neural nets.[63]

One reason there is uncertainty about who attended the event is that McCarthy did not prepare a report on it for the

Rockefeller Foundation. He submitted only a long paper written by Solomonoff on inductive inference.[64] In a letter promising to write a report, McCarthy told the foundation that at the session on automata at the IRE Symposium on Information Theory, to be held in September, ''all of the speakers [Simon, Newell, Rochester, Minsky, Solomonoff, and McCarthy] will be participants in our summer project.'' Because they had missed a deadline, only the papers by Newell and Simon (on the Logic Theorist) and by Rochester and his group (on neural net modeling) would appear in the symposium's proceedings.[65]

Simon recalls some ''tough negotiations'' about the session. McCarthy wanted to chair it and summarize the research done at Dartmouth. ''And we [Simon and Newell] allowed as how that wasn't going to happen and so poor Walter Rosenblith [a colleague of Wiener's at MIT] . . . who was supposed to chair the session, walked around with us around the M.I.T. campus, we strolled down the Memorial Boulevard and so on, negotiating this . . . And finally it was agreed that John McCarthy would get up and give a general speech about what went on and then Al [Newell] would present our work in particular.'' Newell and Simon were satisfied with the arrangements but felt the symposium's audience was not ready to understand their information-processing approach to AI. McCarthy later admitted that Newell and Simon ''felt, perhaps quite correctly, that the situation was anomalous.''[66] Shannon had stopped doing research on automata at this time and gave a paper at the symposium on coding for a noisy channel, which became a classic in the booming field of information theory.[67]

The Rockefeller Foundation was unsure at the time whether the Dartmouth conference had been a success. It probably liked the fiscal responsibility shown by the organizers and Dartmouth College, which returned almost $1,700 of the $7,500 grant as unspent funds.[68] But the foundation was dissatisfied with the lack of a report. When Selfridge spoke to Morison in mid-1957 about the possibility of the foundation providing travel money for US researchers to attend a conference on AI at the National Physical Laboratory in Teddington, Britain, with Selfridge and Minsky as the US organizers, Morison had misgivings. As recorded in his diary, he told Selfridge, ''there has been a little feeling here that we were not entirely well informed as to what went on at the conference held in Hanover [New Hampshire] last summer.'' Yet Morison came away from the interview with an ''excellent impression of S. [Selfridge], and there is no doubt about the qualifications of most of the other people who would be invited to the [Teddington] conference.'' The list included Minsky and McCarthy, as well as such cybernetics researchers as Pitts and Jerome Lettvin from McCulloch's laboratory at MIT and Ashby and Uttley in Britain. Shannon and MacKay were on the advisory committee. Morison now acknowledged that the ''field is an important one and [is] developing very rapidly.''[69] But a month later, in June 1957, the Foundation said it needed more information before funding the conference.[70]

Held in Teddington in late 1958, the conference on ''Mechanisation of Thought Processes'' marked a milestone in the history of AI.[71] Organized by Uttley in Britain and Selfridge in the US, the conference gathered together proponents of the main approaches to machine intelligence of the day. Papers were given by Minsky, McCarthy, and Selfridge on symbolic AI; by McCulloch, Uttley, and Frank Rosenblatt of Cornell University on neural nets; and by Ashby, MacKay, and British cybernetician Gordon Pask on non-neural-net self-organizing systems. Minsky alluded to his conversion to the symbolic method by noting that his proposed heuristic computer program for solving problems in plane geometry was based on notes he had circulated at the Dartmouth conference.[72]

Discussion

What do we learn from this detailed history, based on unpublished archives, about the relationships among cybernetics, automata studies, and the Dartmouth conference on AI?

First of all, the correspondence between Shannon and McCarthy shows that Shannon was not as conservative as McCarthy recalled. Shannon shared McCarthy's goal of creating a science of intelligent machines, but he held a broader view of viable approaches in automata studies, which included neural nets, Turing machines, cybernetic mechanisms, and symbolic processing by computer. Yet it is clear from the correspondence that McCarthy's dissatisfaction with the papers in *Automata Studies* led him to ask the Rockefeller Foundation to fund a conference on AI.

Second, the exchange between Simon and Shannon reveals a sharp instance of how

disciplinary boundaries were increasingly being drawn between brain modeling and what became known as symbolic AI. Shannon pushed ''non-numerical [i.e., symbolic] programming''—by writing an early article on programming a computer to play chess and supervising a graduate student's thesis on a theorem-proving program—but at Dartmouth he proposed to apply his research on information theory to brain modeling and then work up to higher thought processes. Simon turned that research program upside down, or right side up from Simon's point of view. Thus, even though Simon had drawn heavily on Ashby's work on brain modeling, he contributed to the split in the machine intelligence field evident in the amended grant proposal sent to the Rockefeller Foundation: between Simon, Newell, and McCarthy on the symbolic side, and Shannon, Minsky, and Rochester on the brain-modeling side.

Third, we gain a better understanding of the Rockefeller Foundation's role in funding the Dartmouth conference. Although the foundation had funded early computer projects and the mathematical-biological research that inspired Wiener's *Cybernetics*, it did not play a leading role in shaping the related new area of AI. Perhaps because Morison handled the Dartmouth grant rather than Weaver, who had managed the foundation's computer projects, the foundation was not keen on funding the new field of ''mathematical models for thought,'' to use Morison's term. The foundation cut the requested funding nearly in half and apparently did not fund travel to the Teddington conference in 1958, though Morison now had a higher opinion of the field. The foundation unsuccessfully tried to influence the direction of research at Dartmouth by separately funding MacKay's travel to the conference.

One can only speculate how the presence of MacKay—an outspoken proponent of the analog, feedback approach to machine intelligence associated with cybernetics—would have changed the dynamics at Dartmouth and its subsequent ''mythology,'' to use Newell's term. The question is especially apt because McCarthy has recalled that,

> As for myself, one of the reasons for inventing the term ''artificial intelligence'' was to escape association with ''cybernetics.'' Its association with analog feedback seemed misguided, and I wished to avoid having either to accept . . . Wiener as a guru or having to argue with him.[73]

MacKay got his chance to debate McCarthy and Minsky at Teddington in 1958. But by then, McCarthy and Minsky had begun to gather ammunition from their research at MIT with which to fight the battle against brain modeling, especially Rosenblatt's perceptrons.[74]

More generally, the meaning of cybernetics, which initially applied to all approaches to machine intelligence, was increasingly reduced in the US to neural nets and brain modeling after the conference at Dartmouth. Furthermore, MacKay and Uttley even became wary of using the term in 1959 because of its use in Britain by an investment scheme involving research on artificial organisms.[75] In 1968, when Minsky credited cybernetics with spawning three branches of machine intelligence, he effectively limited ''cybernetics'' to the brain-modeling work on self-organizing systems conducted by members of the Ratio Club in Britain and the Biological Computer Laboratory at the University of Illinois. Minsky's genealogy helped mask the pervasiveness of cybernetics in these formative years, until the ''revival'' of neural nets in the 1980s renewed the interest of historians in the place of cybernetics in the early history of AI, and the place of AI in the history of early cybernetics.[76]

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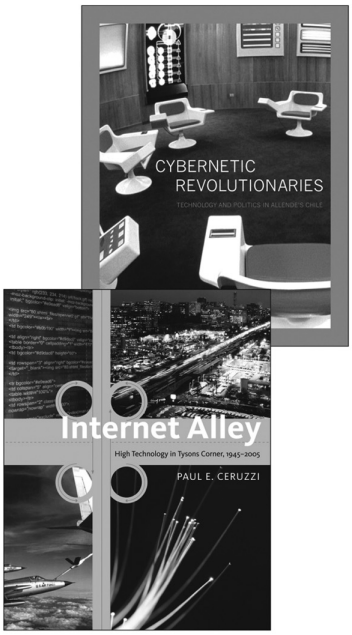

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